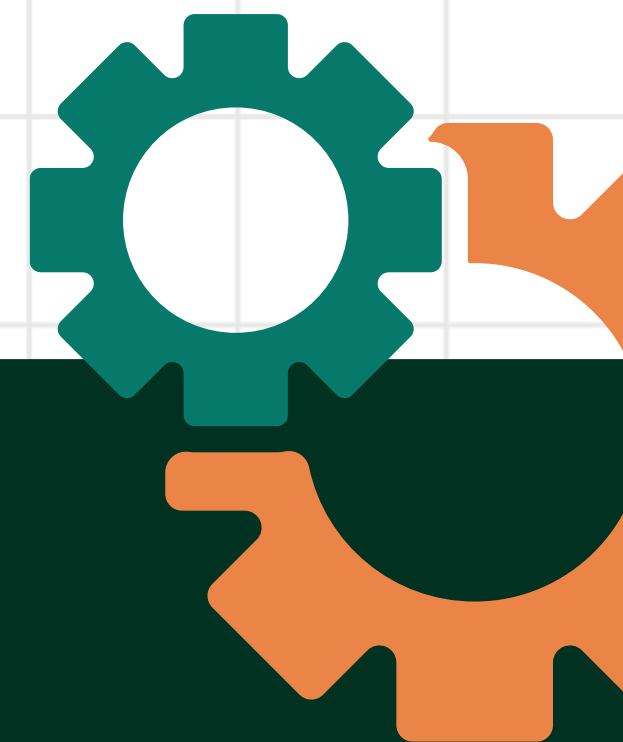


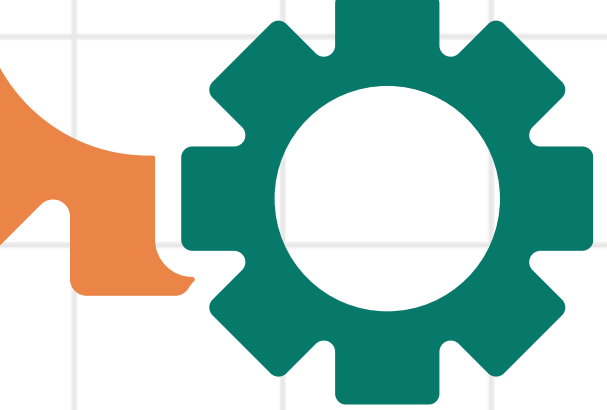


GENERAL SCIENCE
WEEK 3

SIMPLE AND COMPOUND MACHINES

NAME OF TEACHER





DAY 3: LEARNING OBJECTIVES

- Describe the importance of efficiency in engineering by explaining how it reduces energy waste and impacts the sustainability of modern technology
- Identify various ways to enhance machine performance, specifically comparing how lubrication, material choice, and structural design work to minimize friction and energy loss.
- Design a prototype that incorporates specific structural enhancements to improve efficiency, serving as a solution-based model for performance task.

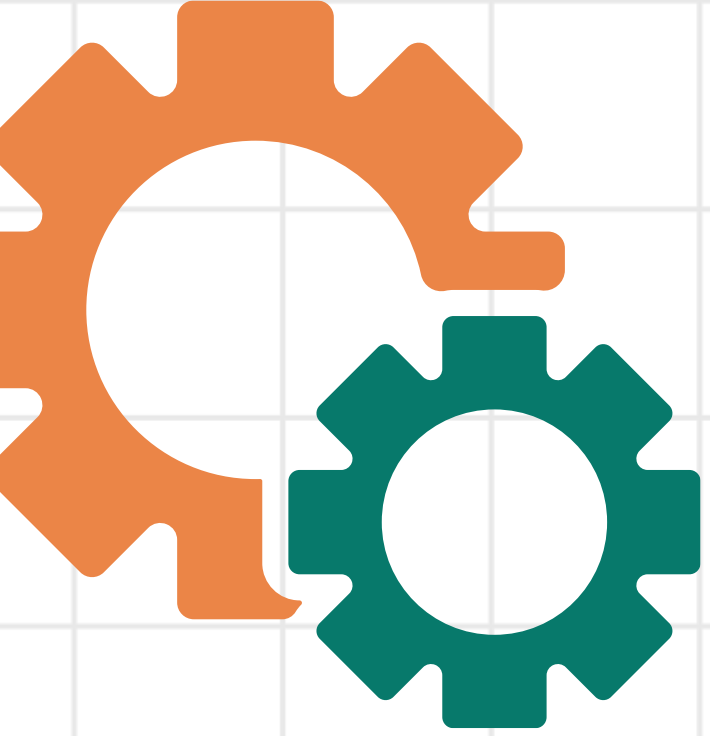
RECAP:

- **Define Efficiency:** In your own words, what does it mean for a machine to be "efficient"?
- **Energy Transformation:** Why is a machine's output energy always less than the input energy? Where does the "lost" energy usually go?

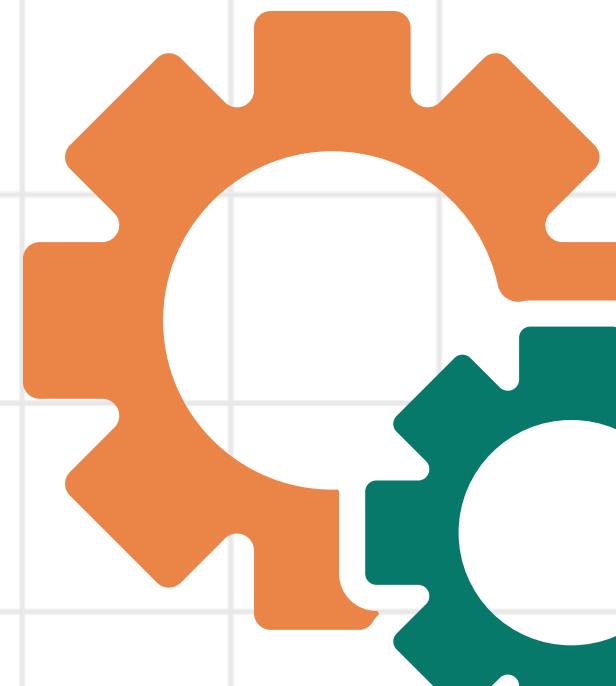
ACTIVITY: MACHINES IN ACTION

1. What do you already know about simple machines?
2. How do you think simple machines help make work easier?
3. Can you name some examples of simple machines you encounter daily?





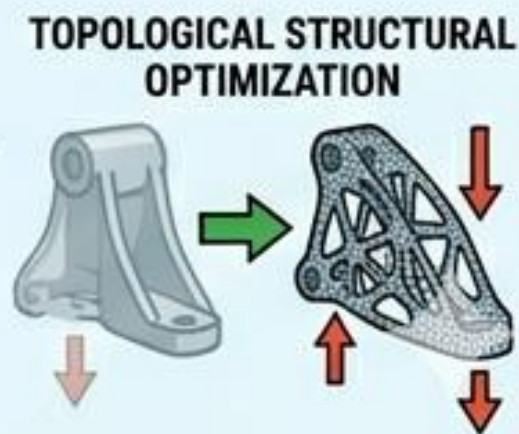
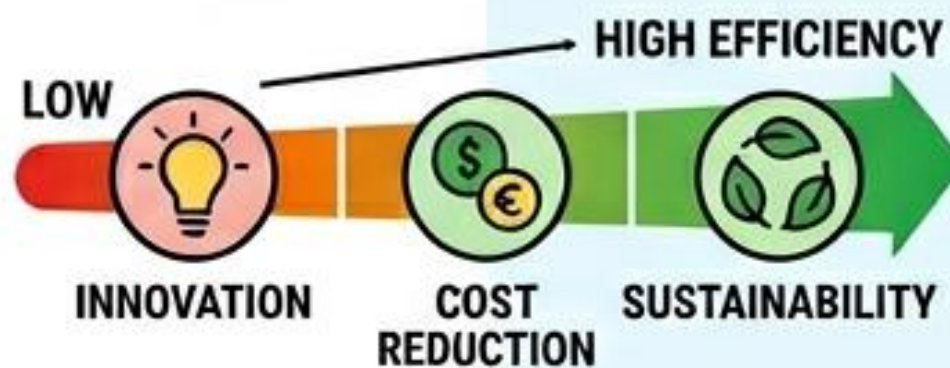
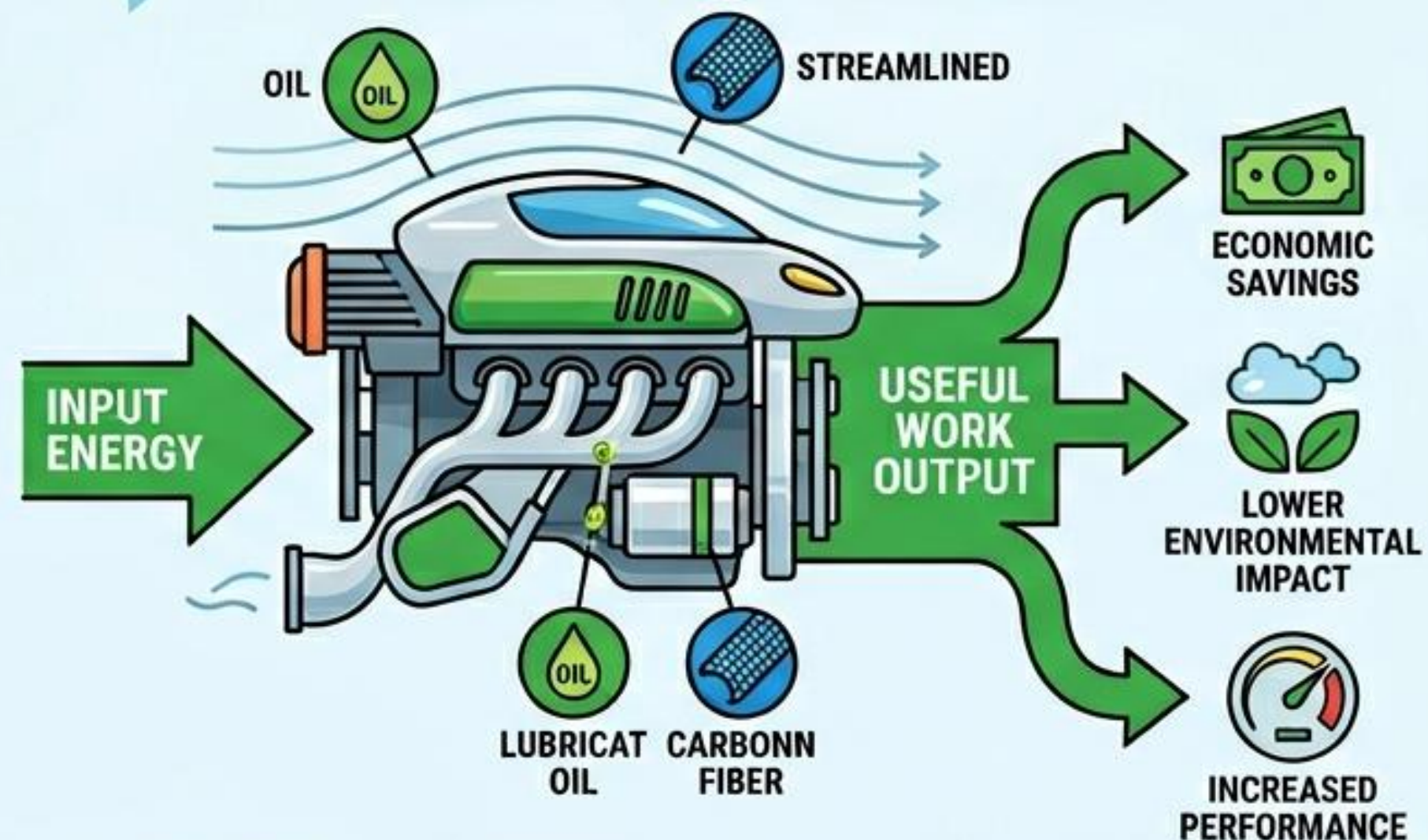
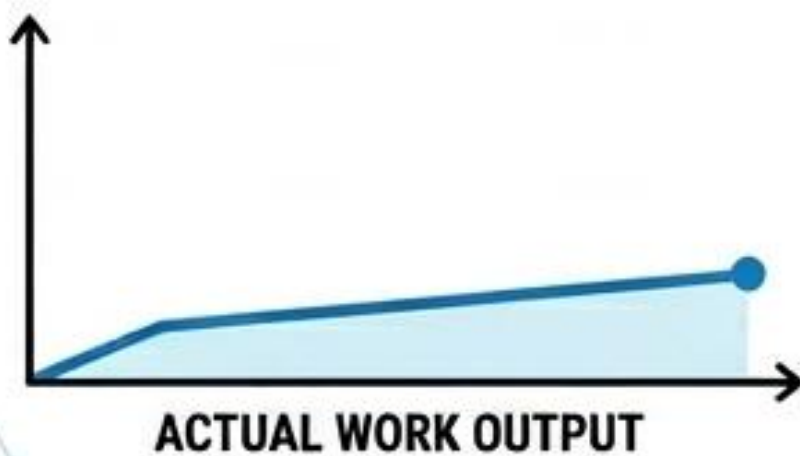
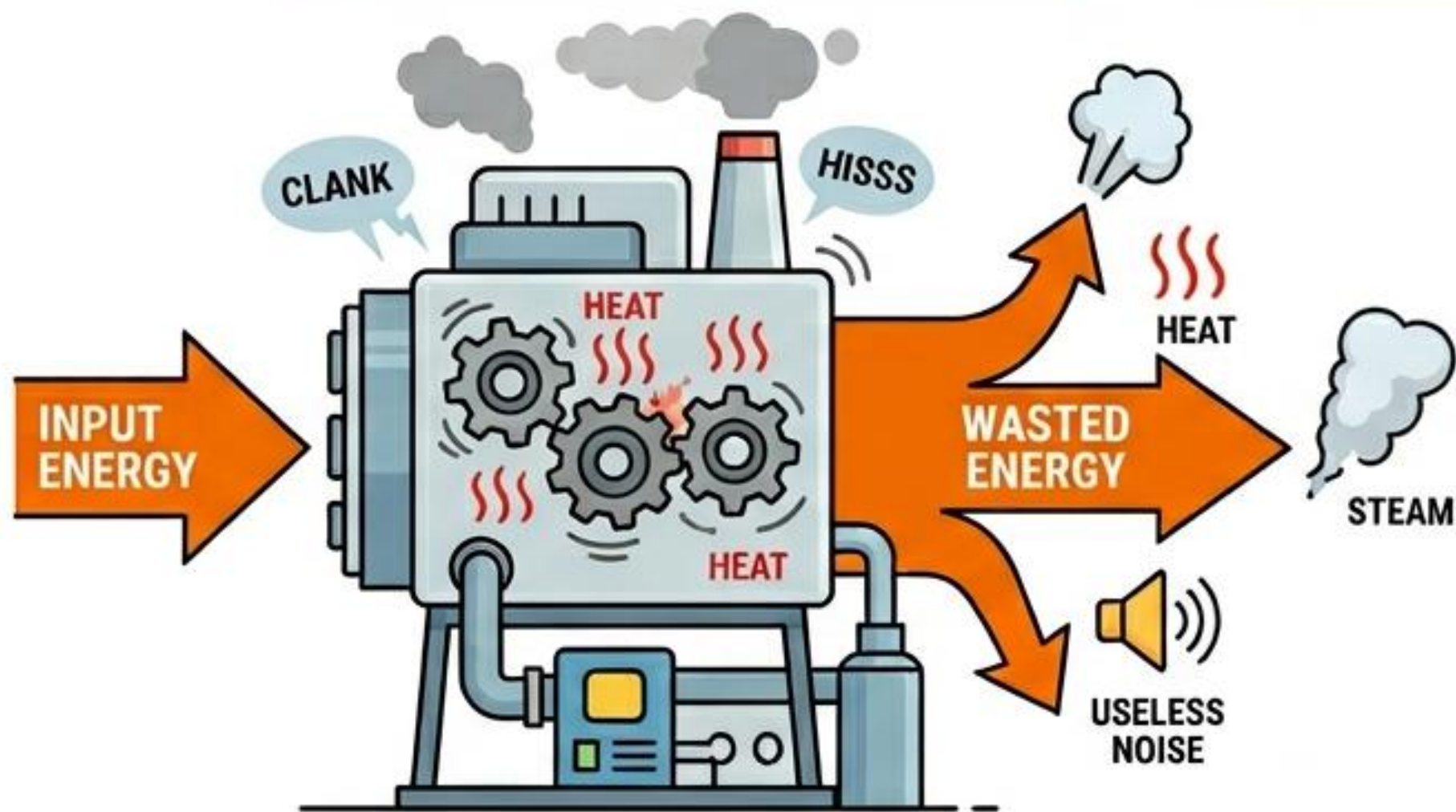
IMPORTANCE OF EFFICIENCY IN ENGINEERING AND TECHNOLOGY



LOW EFFICIENCY SYSTEM

Energy Flow

HIGH EFFICIENCY SYSTEM



Efficiency in engineering isn't just about saving money; it's about doing more with less. In a world with finite resources, engineering high-efficiency systems is the primary way we balance technological progress with environmental survival

1. Energy-Efficient Buildings

Buildings are some of the largest energy consumers globally. By focusing on efficiency, engineers can dramatically reduce a structure's carbon footprint.

- **Thermal Insulation:** Using advanced materials in walls and windows to keep heat in during winter and out during summer.
- **Smart HVAC & Lighting:** Systems that use sensors to adjust temperature and light based on occupancy.
- **The Result:** A significant drop in electricity consumption and lower greenhouse gas emissions.



2. Fuel-Efficient Transportation

In transportation, efficiency is measured by how much "work" (distance/cargo) is done per unit of fuel.

- **Aerodynamics:** Utilizing structural design enhancements to reduce air resistance.
- **Lightweighting:** Replacing heavy steel with aluminum or carbon fiber (material choice) to reduce the energy required to accelerate.
- **The Result:** Improved vehicle performance, longer range for EVs, and lower fuel costs for consumers



3. Automation & AI in Manufacturing

Modern factories use Artificial Intelligence and robotic automation to refine the production process. This is the "Efficiency of Operation."

- **Precision & Accuracy:** Robots can perform tasks with sub-millimeter precision, virtually eliminating "human error" that leads to scrapped parts.
- **Waste Reduction:** AI algorithms calculate the exact amount of raw material needed, ensuring minimal leftover waste.
- **The Result:** Faster production speeds and a massive reduction in industrial waste, making the manufacturing cycle more sustainable.



SECTOR	EFFICIENCY STRATEGY	PRIMARY BENEFIT
Buildings	Smart insulation & HVAC	Reduced electricity & CO2
Transport	Streamlining & lightweight materials	Lower fuel cost & better speed
Manufacturing	AI-driven robotics	Higher accuracy & less water

EFFICIENCY COMPARISON

Ways to Boost Efficiency

To make a machine efficient, engineers focus on three main pillars. As we go through these, think about which one you will prioritize for your prototype performance task.

1. Reducing Friction: Lubrication

Friction represents the primary impediment to mechanical efficiency. When two surfaces engage in kinetic contact, the resulting resistance causes a parasitic conversion of energy into thermal output rather than useful work.

- **How it works:** Lubricants (like oil, grease, or even compressed air) create a microscopic "cushion" between moving parts.
- **Engineering Impact:** Less heat means parts last longer and require less force to move.



2. Reducing Mass: Material Choice

Every extra kilogram a machine weighs requires more energy to move. This is known as overcoming inertia.

- **How it works:** Engineers replace heavy parts (like steel) with modern, lightweight materials like carbon fiber, titanium, or high-strength plastics.
- **Engineering Impact:** A lighter machine uses less fuel and can accelerate or stop much faster.



Carbon Fiber Bike

3. Reducing Drag: Structural Design

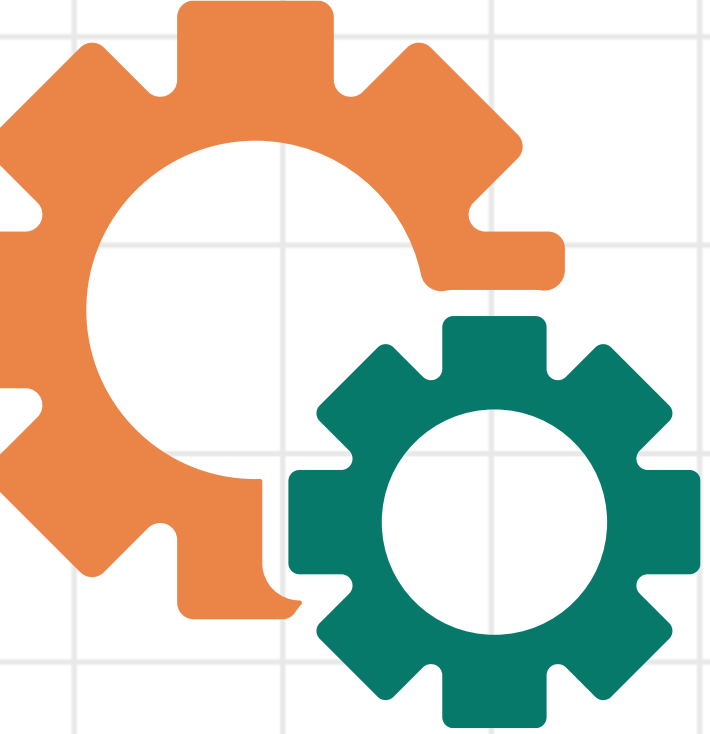
Efficiency is also about how a machine interacts with its environment, especially air or water.

- **How it works:** Streamlining involves shaping a machine to let air flow smoothly around it rather than hitting it like a wall.
- **Engineering Impact:** Proper geometry reduces "drag," allowing the machine to maintain speed without needing extra power.



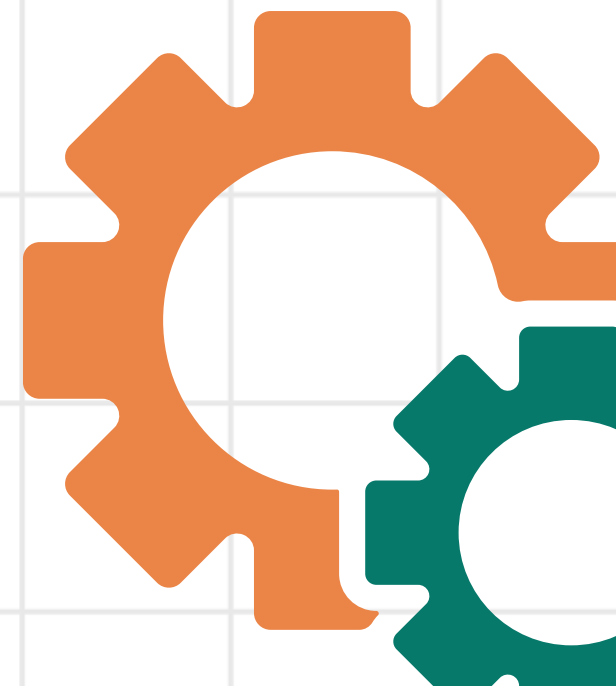
For your performance task, you are expected to propose a design. To start, you need to pick your "Efficiency Focus:

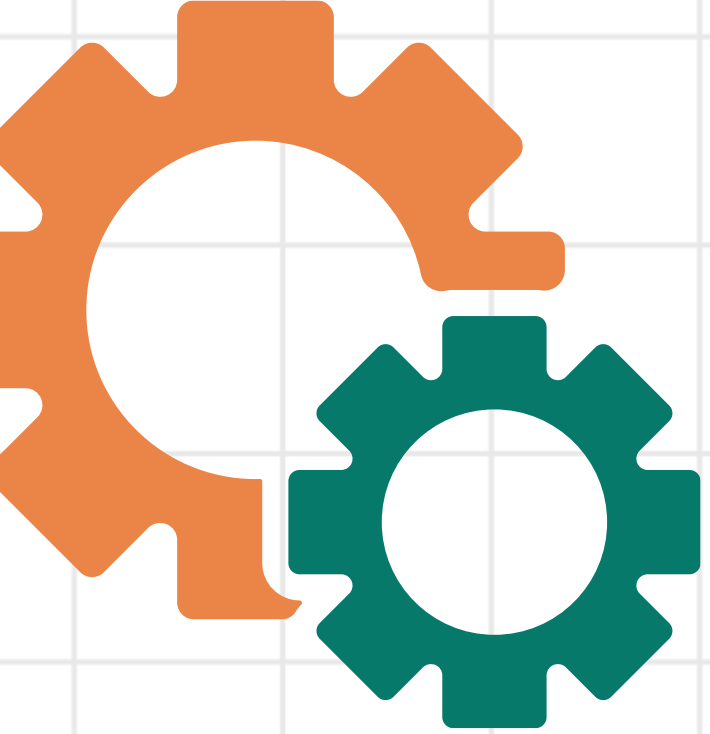
If you want to...	Focus on...
Stop parts from wearing out and getting hot	Lubrication
Make the machine easier to lift or move	Material Choice
Help the machine move faster through air/water	Structural Design



QUIZ

1. Which combination of simple machines is found in a pair of scissors?
 - A. Lever and wedge
 - B. Pulley and inclined plane
 - C. Wheel and axle and screw
 - D. Wedge and inclined plane

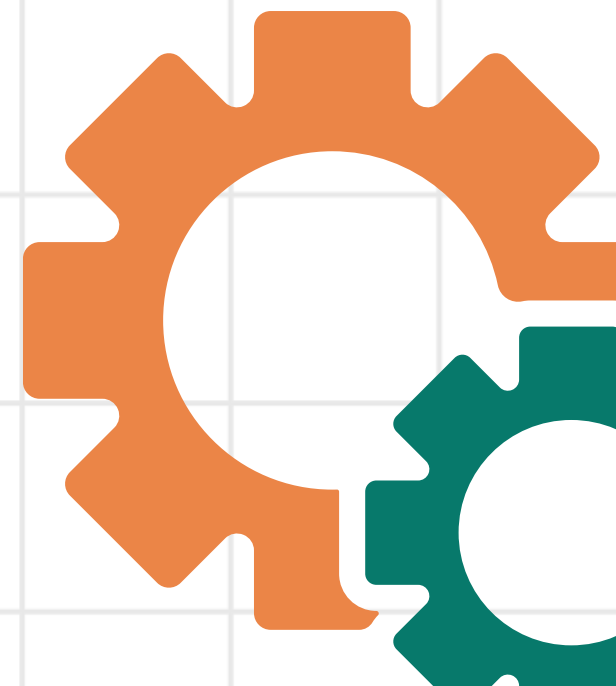


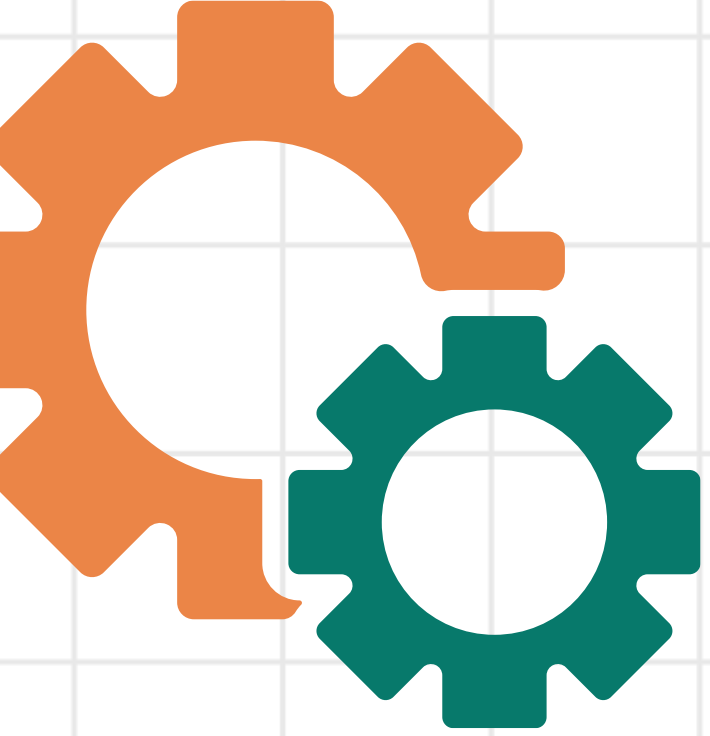


QUIZ

2. Which simple machine is most efficient for lifting heavy objects vertically?

- A. Inclined plane**
- B. Pulley system**
- C. Lever**
- D. Wheel and axle**

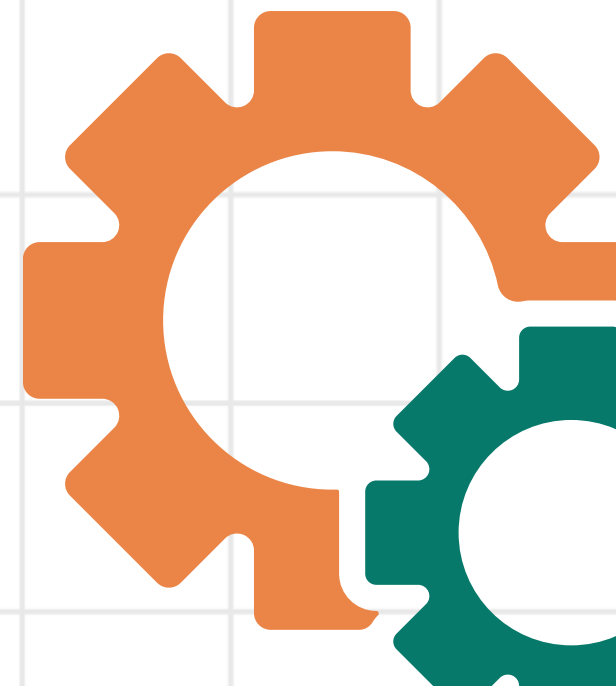


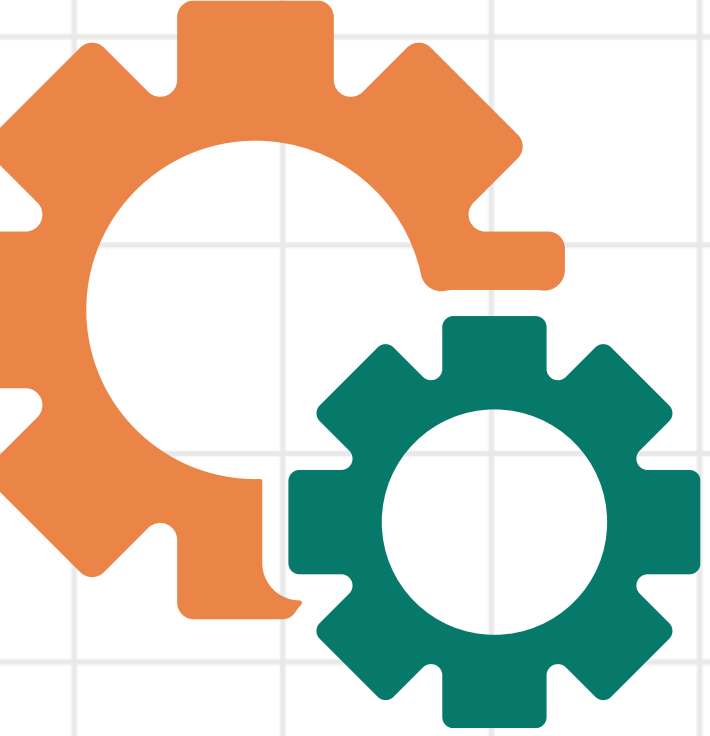


QUIZ

3. Which principle best explains how a lever reduces the effort needed to lift a heavy object?

- A. Mechanical advantage**
- B. Friction reduction**
- C. Energy conservation**
- D. Rotational motion**

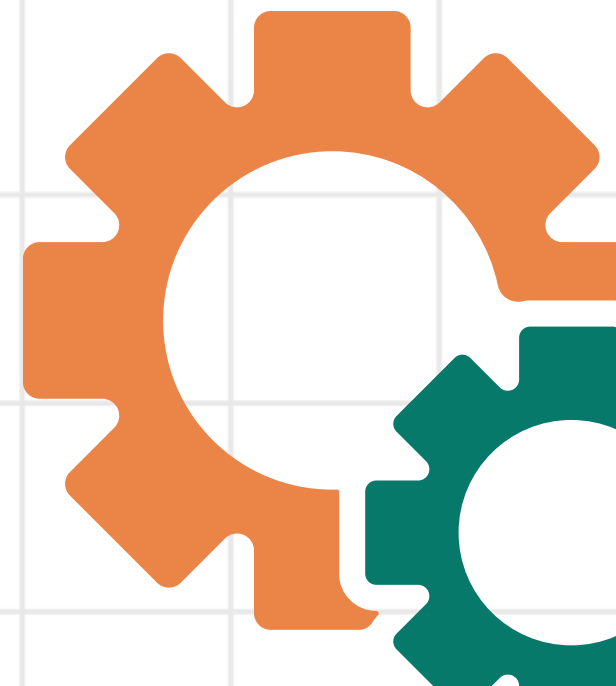


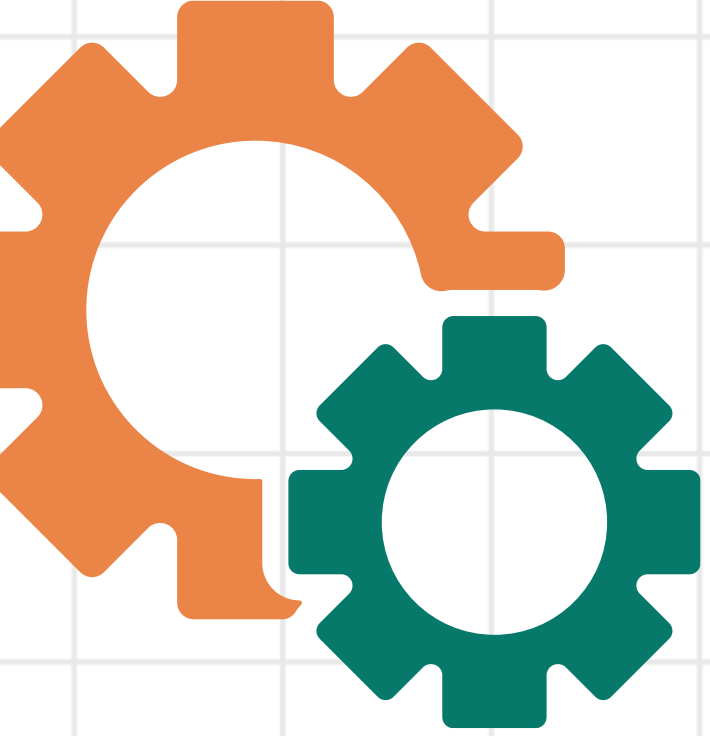


QUIZ

4. Why is lubrication important in improving the efficiency of machines?

- A. It increases mechanical advantage.**
- B. It reduces energy loss due to friction.**
- C. It eliminates the need for input work.**
- D. It increases the speed of motion.**

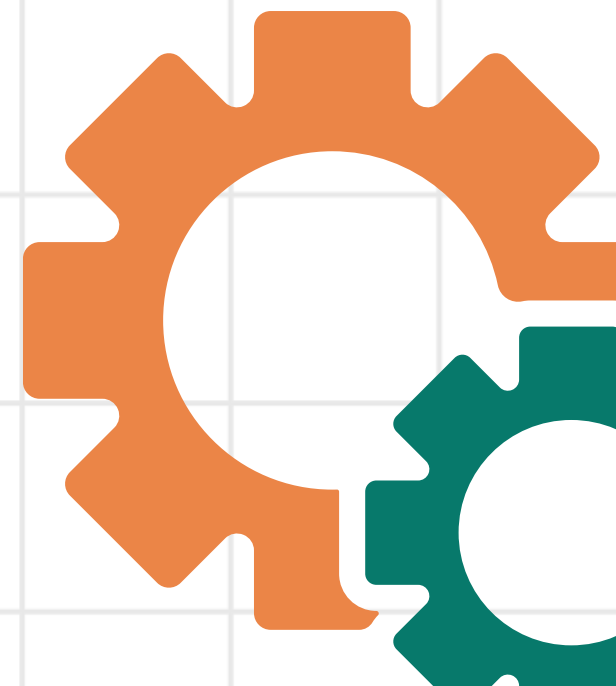


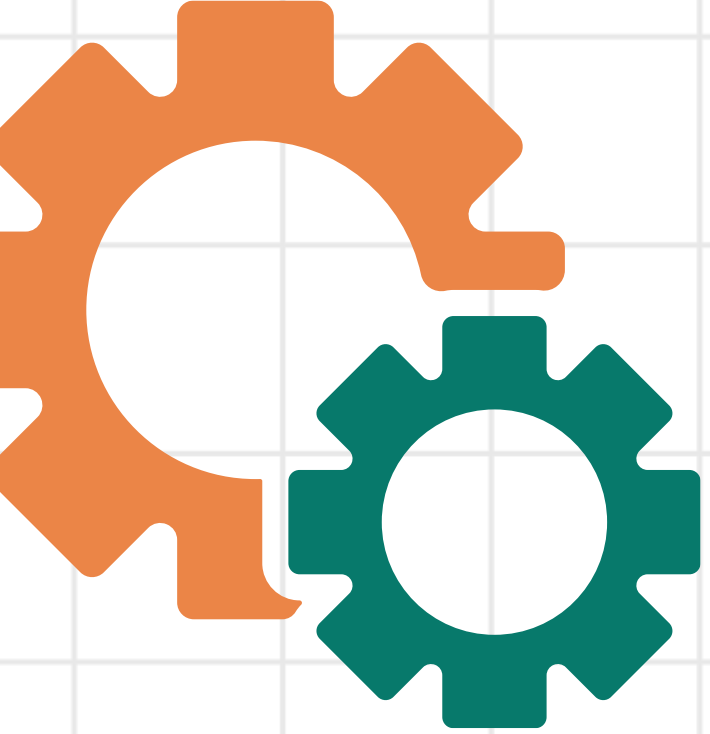


QUIZ

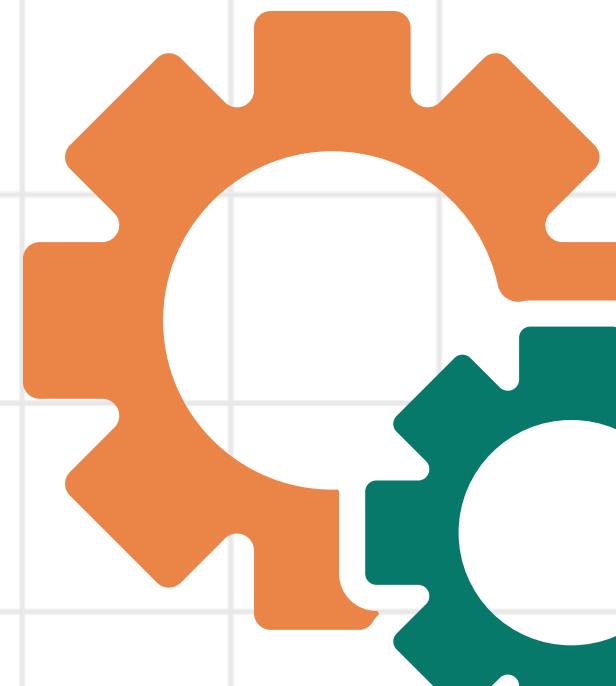
5. What is the primary reason compound machines are more efficient than simple machines?

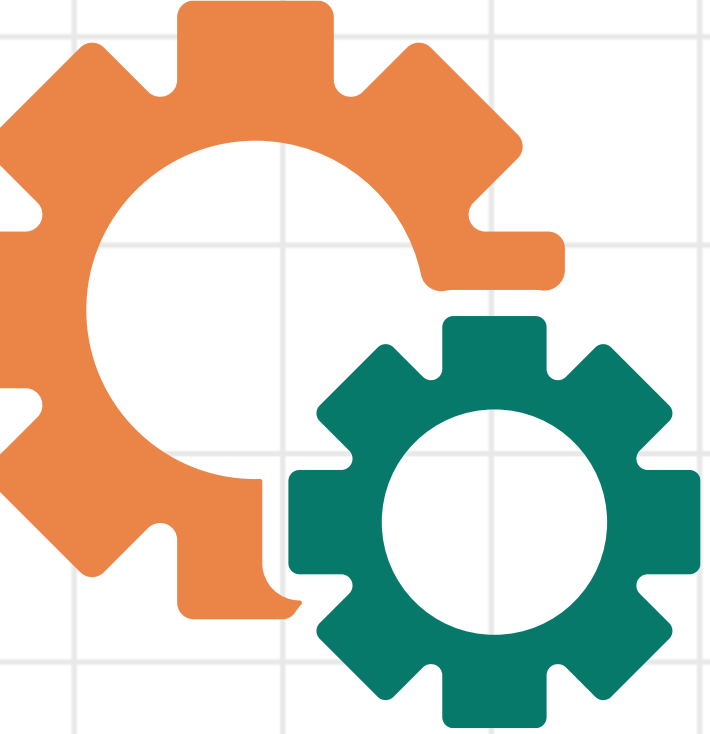
- A. They use less energy.**
- B. They combine multiple functions to reduce effort.**
- C. They eliminate friction completely.**
- D. They require no input work.**





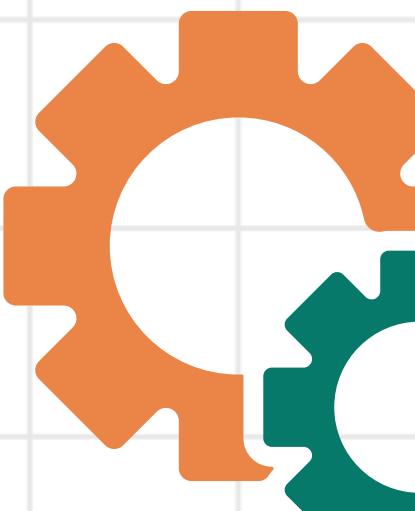
PERFORMANCE TASK
ENGINEERING MOTION: BUILDING
EFFICIENT MACHINES

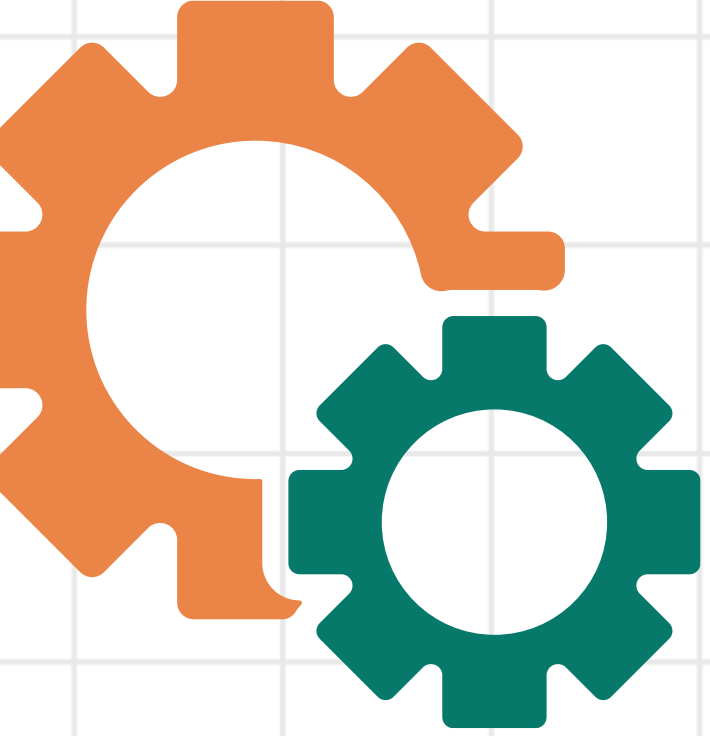




Performance Task: Engineering Motion: Building Efficient Machines

Work in small groups to design, build, and test a working prototype of a simple or compound machine that demonstrates both translational and rotational motion. The group will explain the physics behind the motion of their machine and reflect on its efficiency.

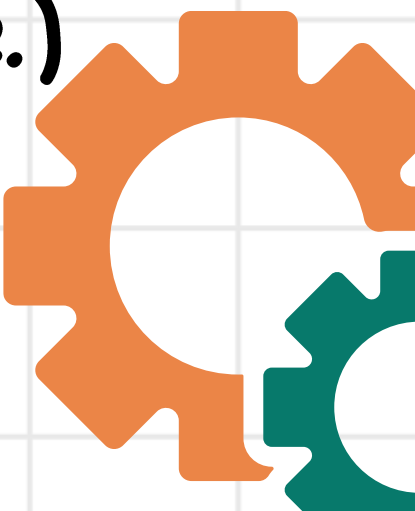




Materials (Suggested)

- **Cardboard, wood, strings, pulleys, rubber bands**
- **Screws, nails, glue, hot glue gun**
- **Plastic wheels, gears, syringes (for hydraulics)**
- **Rulers, protractors, stopwatch, weights**

(You are encouraged to reuse/recycle materials as much as possible.)



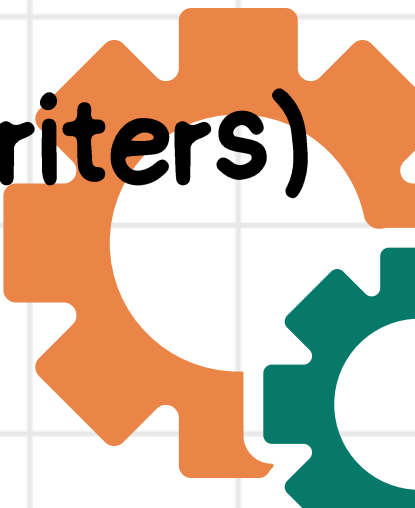
Before the Activity

1. Review the Concepts:

- o Quick recap of rotational motion
- o Quick recap of translational motion
- o Introduce types of simple machines (lever, pulley, inclined plane, wheel and axle, wedge, screw)
- o Explain compound machines and examples

2. Form Student Groups:

- o Assign or let students form groups of 3–5
- o Each group should have mixed strengths (builders, thinkers, writers)
- o Provide guidelines



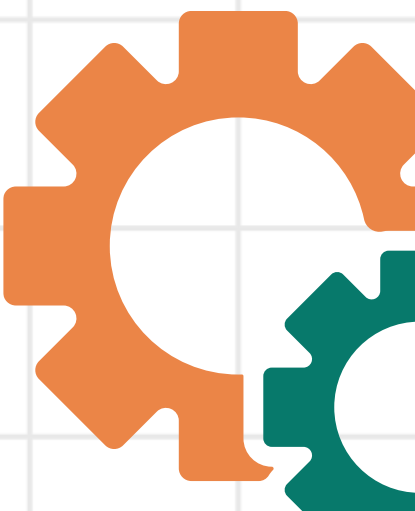
During the Activity

- Assigned as a project over (1–2 weeks):
- Allow time for designing, building, and testing
- Encourage iteration (test, observe, improve)

After the Activity

Each group submits:

- Prototype (demonstrated in class or through video)
- Written Report
- Presentation



Requirements

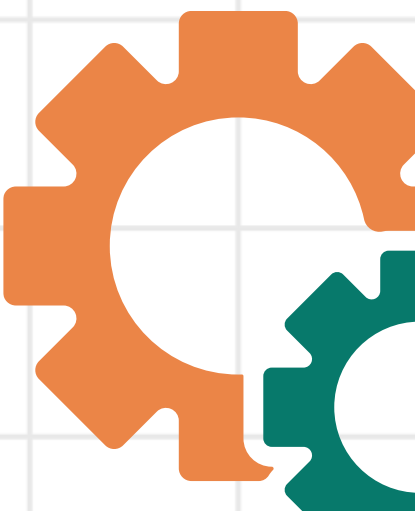
1. Must include:

- At least one rotating component (wheel, gear, pulley, etc.)
- At least one translational component (sliding, pushing, lifting, etc.)
- At least one simple machine (lever, pulley, etc.)

2. Must accomplish a useful task, such as:

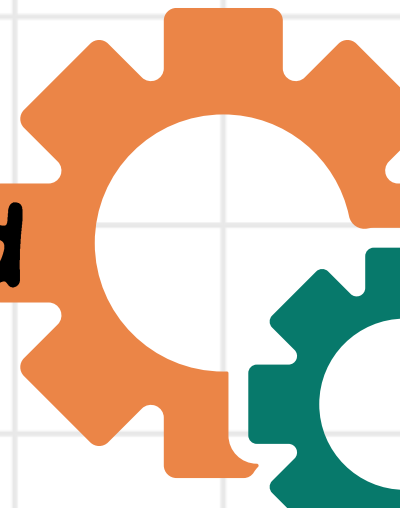
- Lifting a weight
- Moving an object
- Launching something

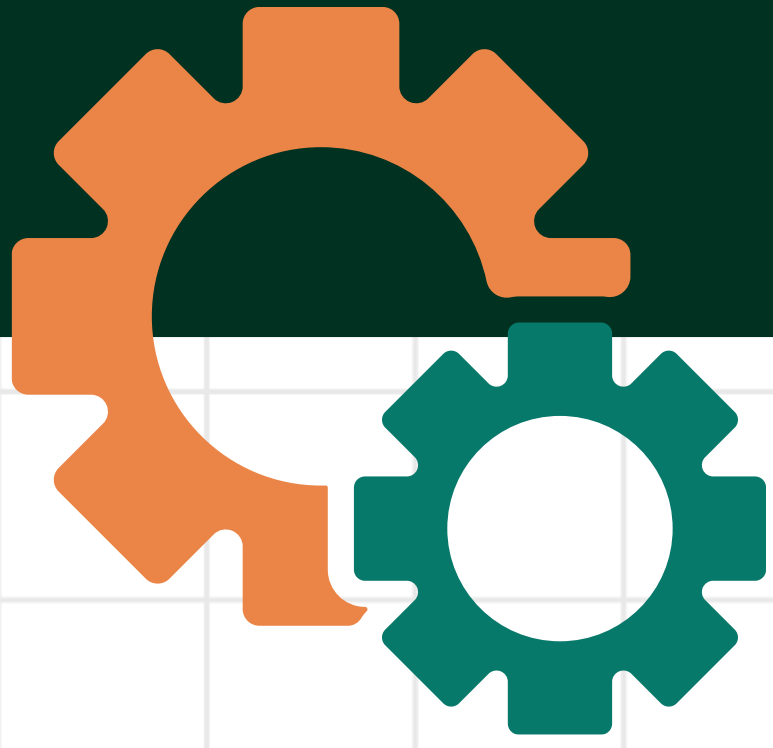
3. Use recycled or low-cost materials



Deliverables

1. Prototype (presented in class or video demonstration)
2. Written Report (2–3 pages) with:
 - Description and labeled diagram of your machine
 - Explanation of how it works using physics terms
 - Calculations of force, torque, efficiency (if applicable)
 - Reflection on design process and teamwork
3. Short Group Presentation (3–5 minutes):
 - Show the working prototype
 - Explain how translational and rotational motion are involved





THANK YOU!

I hope you learn something new today.

